

Exercise 4.1. Let n and $m \neq 0$ be two natural numbers. Show that there exist unique natural numbers a and b such that $n = am + b$ with $b < m$.

You can try to [verify this exercise with Lean](#).

Hints for Lean

```
import Mathlib

theorem euclidean_division (n m : Nat) (hm : m ≠ 0) :
  ∃! a : Nat, ∃! b : Nat, n = a * m + b ∧ b < m := by
  have hm' : m - 1 + 1 = m := by
    apply Nat.sub_add_cancel
    apply Nat.succ_le_of_lt
    exact Nat.pos_of_ne_zero hm
  have hm_le : m - 1 < m := by
    rw [← hm']
    exact Nat.lt_succ_self (m - 1)
  induction n with
  | zero =>
    -- Proof when n = 0.
    sorry
  | succ n ih =>
    -- Proof for n + 1, assuming the result is true for n.
    sorry
```

When dealing with natural numbers, Lean uses [truncated subtraction](#). So for example $3 - 5 = 0$. This means that the claims `hm'` and `hm_le` above are not automatic. In particular, a tactic like `ring` will not be able to prove `hm'`. The [Natural Number Game](#) may be useful to learn more about natural numbers in Lean.

A [solution](#) is available on the repository

<https://github.com/pedro-nunez/lean-algebra-2021>.

[†]Exercises designed for the lecture [Algebra and Number Theory](#), for which I was a teaching assistant. This was an introductory lecture on abstract algebra, polynomial rings and Galois theory, taught by Annette Huber-Klawitter at the University of Freiburg during the Winter Semester 2021/2022.